

MENTORA ACADEMY

DEPARTMENT OF ADVANCED CHEMISTRY · CAMBRIDGE INTERNATIONAL A LEVEL (9701)

Physical Chemistry Practical: Paper 5 Master Suite

10 Most Frequently Tested Planning, Analysis & Evaluation Questions (Past 5 Years Analysis: 2020–2024)

CANDIDATE NAME:	Urwah	CENTRE NUMBER:	PK082	CANDIDATE NO:	0142
COMPONENT:	Paper 5 — Planning, Analysis and Evaluation (Past 10 Years Archive)	DURATION:	2 Hours 30 Minutes	MAX MARKS:	100 Marks

READ THESE INSTRUCTIONS FIRST

- Write your name, Centre number, and candidate number in the boxes above.
- Answer all questions in the spaces provided on the question paper.
- In Question 1 (Planning): define independent, dependent, and controlled variables; describe step-by-step procedures with apparatus diagrams; calculate required quantities; state hazard risk assessments.
- In Question 2 (Analysis, Conclusions & Evaluation): process raw experimental data into derived columns; quote values to consistent significant figures; plot graphs with appropriate linear scales; determine gradients and intercepts; identify anomalous points; evaluate uncertainties.
- You may use an approved scientific calculator. The Periodic Table is provided as standard reference.

EXAMINATION BREAKDOWN & SYLLABUS SKILLS TESTED

Question	Investigation Title & Focus	Syllabus Ref	Type of Task	Marks
Question 1	Planning an Electrochemical Determination of the Faraday Constant	9701/52/M/J/23/Q1	PLANNING	10
Question 2	Analysis and Graphical Evaluation of Persulfate–Iodide Reaction Kinetics	9701/51/O/N/23/Q2	ANALYSIS & EVALUATION	10
Question 3	Planning Activation Energy Determination via Arrhenius Temperature Kinetics	9701/52/M/J/22/Q1	PLANNING	10
Question 4	Analysis and Evaluation of the Solubility Product (K_{sp}) of Calcium Hydroxide	9701/53/O/N/22/Q2	ANALYSIS & EVALUATION	10
Question 5	Planning Calorimetric Determination of the Hydration Enthalpy via Hess's Law	9701/51/M/J/24/Q1	PLANNING	10
Question 6	Analysis and Evaluation of the Catalytic Decomposition of Hydrogen Peroxide	9701/52/O/N/21/Q2	ANALYSIS & EVALUATION	10
Question 7	Planning the Determination of Partition Coefficient (K_{pc}) of Ethanoic Acid	9701/53/M/J/21/Q1	PLANNING	10
Question 8	Quantitative Analysis of Cell Potentials and Verification of the Nernst Equation	9701/51/O/N/24/Q2	ANALYSIS & EVALUATION	10
Question 9	Planning the Determination of Molar Mass of a Volatile Liquid via the Dumas Method	9701/52/M/J/20/Q1	PLANNING	10
Question 10	Analysis of Apparatus Errors, Anomalies, and Standard Deviation in Calorimetry	9701/51/M/J/23/Q2	ANALYSIS & EVALUATION	10

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QUESTION 1

PLANNING

[Total: 10 Marks]

Planning an Electrochemical Determination of the Faraday Constant

Cambridge Syllabus Reference: 9701/52/M/J/23/Q1

A student plans an experiment to determine the Faraday constant, F , by the electrolysis of aqueous copper(II) sulfate using weighed copper electrodes.

The half-equation at the cathode is: $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$.

The relationship is given by: $m = (M \times I \times t) / (z \times F)$, where m is mass of copper deposited, $M = 63.5 \text{ g mol}^{-1}$, I is current, t is time in seconds, and $z = 2$.

(a) Draw a fully labelled diagram of the complete electrical circuit and electrolytic cell required for this experiment. [3]

(b) Identify the independent variable, dependent variable, and one variable that must be kept constant. [3]

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(c) State the procedure to clean and dry the cathode before and after electrolysis to ensure mass accuracy. [2]

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(d) Explain how the student should use the measured values to calculate an experimental value for F . [2]

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QUESTION 2

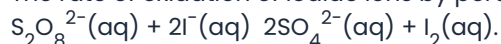
ANALYSIS & EVALUATION

[Total: 10 Marks]

Analysis and Graphical Evaluation of Persulfate–Iodide Reaction Kinetics

Cambridge Syllabus Reference: 9701/51/O/N/23/Q2

The rate of oxidation of iodide ions by peroxodisulfate ions was investigated using the iodine clock method:



A small fixed amount of sodium thiosulfate and starch indicator were added. The time t taken for the blue-black color to appear was recorded at various concentrations of $\text{S}_2\text{O}_8^{2-}$, with $[\text{I}^-]$ kept in large excess.

- (a) Explain why $1/t$ is directly proportional to the initial rate of reaction. [2]
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- (b) The student plotted $\log(1/t)$ on the y-axis against $\log[\text{S}_2\text{O}_8^{2-}]$ on the x-axis. State how the order of reaction with respect to $\text{S}_2\text{O}_8^{2-}$ is determined from this graph. [2]
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- (c) The gradient of the line was found to be +0.97. Deduce the order with respect to $\text{S}_2\text{O}_8^{2-}$ and explain the significance of the 0.03 deviation. [2]
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- (d) Identify one anomalous result on the student's graph and suggest a specific experimental error that could have caused the blue-black color to appear too slowly. [2]
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- (e) Suggest how the precision of timing the endpoint could be significantly improved. [2]
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QUESTION 3

PLANNING

[Total: 10 Marks]

Planning Activation Energy Determination via Arrhenius Temperature Kinetics

Cambridge Syllabus Reference: 9701/52/M/J/22/Q1

The Arrhenius equation describes the variation of rate constant k with absolute temperature T :

$$\ln k = \ln A - (E_a / RT).$$

For a reaction where initial rate $1/t$, the equation can be written as: $\ln(1/t) = -(E_a / R)(1/T) + \text{constant}$.

Plan an investigation to determine the activation energy, E_a , of the reaction between sodium thiosulfate and hydrochloric acid over the temperature range 20 °C to 60 °C.

- (a) Specify the apparatus and temperature control method to ensure thermal equilibrium between reactants before mixing. [3]

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- (b) Describe the step-by-step procedure to measure the reaction rate proxy at each temperature. [3]

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- (c) State the columns that must be included in the student's processed data table. [2]

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- (d) Explain how E_a is calculated from the gradient of the graph of $\ln(1/t)$ against $1/T$. [2]

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QUESTION 4

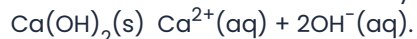
ANALYSIS & EVALUATION

[Total: 10 Marks]

Analysis and Evaluation of the Solubility Product (K_{sp}) of Calcium Hydroxide

Cambridge Syllabus Reference: 9701/53/O/N/22/Q2

A saturated solution of calcium hydroxide was prepared at 298 K according to the equilibrium:



The solubility product expression is: $K_{sp} = [\text{Ca}^{2+}][\text{OH}^{-}]^2$.

A 25.0 cm^3 portion of the filtered saturated solution was titrated against $0.0500 \text{ mol dm}^{-3}$ HCl using phenolphthalein indicator.

- (a) Explain why the saturated solution must be filtered before titration, and state a precaution during filtration. [2]

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- (b) If the mean titre of $0.0500 \text{ mol dm}^{-3}$ HCl is 17.40 cm^3 , calculate $[\text{OH}^{-}]$ and $[\text{Ca}^{2+}]$ in the saturated solution. [3]

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- (c) Calculate the numerical value of K_{sp} for Ca(OH)_2 at 298 K and state its units. [3]

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- (d) Predict and explain the effect on the calculated K_{sp} if the solution absorbed atmospheric CO_2 prior to titration. [2]

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QUESTION 5

PLANNING

[Total: 10 Marks]

Planning Calorimetric Determination of the Hydration Enthalpy via Hess's Law

Cambridge Syllabus Reference: 9701/51/M/J/24/Q1

Hydrated magnesium chloride, $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$, cannot have its enthalpy of hydration measured directly by calorimetry.

However, the enthalpy of solution of anhydrous $\text{MgCl}_2(\text{s})$ (H_1) and the enthalpy of solution of hydrated $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}(\text{s})$ (H_2) can both be measured.

Plan two calorimetry experiments using a simple polystyrene cup calorimeter to determine H_1 and H_2 .

(a) Draw a Hess's Law energy cycle connecting anhydrous $\text{MgCl}_2(\text{s})$, hydrated $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}(\text{s})$, and $\text{MgCl}_2(\text{aq})$. [3]

(b) State the mass of water and mass of salts to use to ensure a measurable temperature change of at least 5°C without exceeding solubility limits. [2]

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(c) Describe how cooling curve graphical extrapolation should be used to correct for heat loss. [3]

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(d) Explain why anhydrous MgCl_2 must be stored in a tightly stoppered container until immediately before weighing. [2]

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QUESTION 6

ANALYSIS & EVALUATION

[Total: 10 Marks]

Analysis and Evaluation of the Catalytic Decomposition of Hydrogen Peroxide

Cambridge Syllabus Reference: 9701/52/O/N/21/Q2

A student investigated the catalytic decomposition of hydrogen peroxide:



The volume of oxygen evolved was collected in a 100 cm³ gas syringe and recorded every 30 seconds.

- (a) Explain how the volume of O₂ evolved at time t , V_t , and the final volume at infinity, V_∞ , are used to find [H₂O₂] remaining. [2]
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- (b) Describe how successive half-lives ($t_{1/2}$) are determined from a graph of ($V_\infty - V_t$) against time. [2]
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- (c) State two reasons why the recorded volume of oxygen gas might be lower than the theoretical volume. [2]
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- (d) Explain the effect of doubling the mass of MnO₂ catalyst on the rate of reaction and on the final volume V_∞ . [2]
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- (e) Suggest how the reaction can be stopped instantly if an aliquot method were used instead of continuous gas collection. [2]
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QUESTION 7

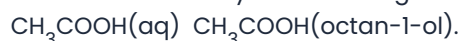
PLANNING

[Total: 10 Marks]

Planning the Determination of Partition Coefficient (K_{pc}) of Ethanoic Acid

Cambridge Syllabus Reference: 9701/53/M/J/21/Q1

When ethanoic acid is shaken with water and an immiscible organic solvent such as octan-1-ol, it distributes between the two layers according to the partition equilibrium:



The partition coefficient is given by: $K_{pc} = [\text{CH}_3\text{COOH}]_{\text{org}} / [\text{CH}_3\text{COOH}]_{\text{aq}}$.

Plan an experiment to determine K_{pc} at room temperature.

- (a) Describe how the two layers should be mixed and brought to equilibrium in a separating funnel. [2]
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- (b) Explain how a sample of the lower aqueous layer should be obtained without contamination by the upper organic layer. [2]
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- (c) State the volumetric titration method to determine the concentration of ethanoic acid in the separated layers. [3]
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- (d) State the safety precautions required when handling octan-1-ol and concentrated ethanoic acid. [2]
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- (e) State the assumption made regarding the molecular state of ethanoic acid in both solvents. [1]
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QUESTION 8

ANALYSIS & EVALUATION

[Total: 10 Marks]

Quantitative Analysis of Cell Potentials and Verification of the Nernst Equation

Cambridge Syllabus Reference: 9701/51/O/N/24/Q2

A student investigated the relationship between copper half-cell potential and $[\text{Cu}^{2+}(\text{aq})]$ using a cell consisting of a Cu^{2+}/Cu half-cell and a standard hydrogen electrode (SHE).

According to the Nernst equation at 298 K: $E = E^{\circ} + (0.059 / z) \log[\text{Cu}^{2+}]$.

The table below shows the measured cell potentials for five different concentrations of Cu^{2+} .

- (a) Complete the table by calculating $\log[\text{Cu}^{2+}]$ to two decimal places.

[2]

$[\text{Cu}^{2+}] / \text{mol dm}^{-3}$	$\log[\text{Cu}^{2+}]$	$E_{\text{cell}} / \text{V}$
1.00	0.00	+0.340
0.10	-1.00	+0.311
0.010	-2.00	+0.281
0.0010	-3.00	+0.250
0.00010	-4.00	+0.222

- (b) State what graph should be plotted to confirm the validity of the Nernst equation.

[2]

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- (c) Calculate the theoretical gradient of this graph for $z = 2$.

[2]

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- (d) Explain how the standard electrode potential $E^{\circ}(\text{Cu}^{2+}/\text{Cu})$ is determined from the graph.

[2]

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- (e) Explain why a high-resistance digital voltmeter must be used to measure cell potential.

[2]

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QUESTION 9

PLANNING

[Total: 10 Marks]

Planning the Determination of Molar Mass of a Volatile Liquid via the Dumas Method

Cambridge Syllabus Reference: 9701/52/M/J/20/Q1

The molar mass of an unknown volatile organic liquid can be determined by vaporising a known mass inside a gas syringe placed inside a temperature-controlled heating jacket or water oven.

The ideal gas equation relates the variables: $pV = (m/M)RT$ $M = mRT / pV$.

Plan an experiment to determine the molar mass of liquid hexane (boiling point 69 °C).

- (a) Specify the apparatus setup to inject a micro-sample of liquid into the gas syringe without gas leakage. [2]

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- (b) Describe the mass measurement technique using weighing by difference. [2]

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- (c) State the temperature at which the syringe oven must be maintained and explain why. [2]

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- (d) List the four experimental measurements required to calculate M. [2]

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- (e) Explain the effect on the calculated molar mass if the vapor does not behave ideally due to significant intermolecular forces. [2]

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QUESTION 10

ANALYSIS & EVALUATION

[Total: 10 Marks]

Analysis of Apparatus Errors, Anomalies, and Standard Deviation in Calorimetry

Cambridge Syllabus Reference: 9701/51/M/J/23/Q2

A student carries out repeated calorimetric runs to determine the enthalpy of neutralization between 25.0 cm^3 of 2.00 mol dm^{-3} NaOH and 25.0 cm^3 of 2.00 mol dm^{-3} HCl.

Apparatus used: 50 cm^3 measuring cylinder ($\pm 0.5 \text{ cm}^3$), thermometer graduated in 0.2°C ($\pm 0.1^\circ\text{C}$).

- (a) Calculate the percentage uncertainty in measuring 25.0 cm^3 of solution with the measuring cylinder. [2]

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- (b) If the temperature rise recorded was $T = 13.4^\circ\text{C}$, calculate the percentage uncertainty in T . [2]

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- (c) Calculate the overall combined apparatus percentage uncertainty for this experiment. [2]

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- (d) The student obtained neutralization enthalpies: -54.2 , -58.1 , -53.9 , $-54.0 \text{ kJ mol}^{-1}$. Identify the anomaly and calculate the mean of concordant values. [2]

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- (e) Suggest two apparatus modifications that would significantly improve the accuracy of the result. [2]

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Comprehensive Examiner Mark Scheme

Marking points, indicative responses, and error-carried-forward (ECF) guidelines

MARK SCHEME: QUESTION 1 (PLANNING) — Planning an Electrochemical Determination of the Faraday Constant

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Direct current power supply connected in series with variable resistor / rheostat, ammeter, and electrolytic cell [1]; copper anode and copper cathode immersed in aqueous CuSO ₄ [1]; cathode clearly labelled as negative electrode where mass increases [1].	[3]
(b)	Independent variable: time of electrolysis <i>t</i> (or current <i>I</i>) [1]; Dependent variable: change in mass of cathode <i>m</i> [1]; Controlled variable: constant current / temperature of solution / surface area of immersion [1].	[3]
(c)	Clean electrode surface with fine emery paper / sand paper before electrolysis to remove oxide layer, rinse with distilled water [1]; after electrolysis, wash gently with propanone / ethanol and allow to dry in a desiccator / gentle warm air stream without dislodging deposit [1].	[2]
(d)	Rearrange formula: $F = \frac{(M \times I \times t)}{(2 \times m)}$ [1]; substitute measured current (<i>A</i>), time (<i>s</i>), and mass gained (<i>g</i>) to evaluate <i>F</i> in C mol ⁻¹ [1].	[2]

MARK SCHEME: QUESTION 2 (ANALYSIS & EVALUATION) — Analysis and Graphical Evaluation of Persulfate–Iodide Reaction Kinetics

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	The amount of iodine produced to react with the fixed amount of thiosulfate is constant ([I ₂] is constant) [1]; initial rate = [I ₂]/ <i>t</i> [1].	[2]
(b)	Rate = $k[S_2O_8^{2-}]^m$ log(rate) = log(<i>k</i>) + <i>m</i> log[S ₂ O ₈ ²⁻] [1]; the order <i>m</i> is equal to the gradient of the best-fit straight line [1].	[2]
(c)	Order is 1 (first order) [1]; deviation of 0.03 is within experimental / timing uncertainty of the clock endpoint [1].	[2]
(d)	Identify point lying below the line of best fit (longer time <i>t</i>) [1]; cause: temperature decreased during run / student inadvertently added extra thiosulfate volume / delayed starch addition [1].	[2]
(e)	Use a colorimeter / spectrophotometer coupled to a datalogger [1]; eliminate subjective visual judgment of colour transition [1].	[2]

MARK SCHEME: QUESTION 3 (PLANNING) — Planning Activation Energy Determination via Arrhenius Temperature Kinetics

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Thermostatically controlled water bath with stirrer [1]; separate boiling tubes / conical flasks containing Na ₂ S ₂ O ₃ and HCl placed in water bath for at least 5 minutes [1]; thermometer to measure temperature of reactant mixture immediately before mixing [1].	[3]
(b)	Place conical flask over printed black cross on white tile [1]; mix equal specified volumes of preheated solutions and immediately start stopwatch [1]; swirl once, view vertically downwards through solution, and stop stopwatch when cross is obscured by precipitated sulfur [1].	[3]
(c)	Columns for: Temperature <i>T</i> in °C, Absolute temperature <i>T</i> in K, 1/ <i>T</i> in K ⁻¹ , Time <i>t</i> in s, 1/ <i>t</i> in s ⁻¹ , and ln(1/ <i>t</i>) [2].	[2]
(d)	Gradient = - <i>E</i> _a / <i>R</i> [1]; <i>E</i> _a = -gradient × 8.314 J K ⁻¹ mol ⁻¹ (convert to kJ mol ⁻¹ by dividing by 1000) [1].	[2]

MARK SCHEME: QUESTION 4 (ANALYSIS & EVALUATION) — Analysis and Evaluation of the Solubility Product (K_{sp}) of Calcium Hydroxide

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Filter to remove undissolved solid Ca(OH) ₂ which would otherwise dissolve during titration and react with acid [1]; discard the first few cm ³ of filtrate because filter paper can adsorb solute ions [1].	[2]
(b)	Moles HCl = 0.0500 × (17.40 / 1000) = 8.70 × 10 ⁻⁴ mol [1]; Moles OH ⁻ in 25.0 cm ³ = 8.70 × 10 ⁻⁴ mol [OH ⁻] = 8.70 × 10 ⁻⁴ / 0.0250 = 0.0348 mol dm ⁻³ [1]; [Ca ²⁺] = 0.5 × [OH ⁻] = 0.0174 mol dm ⁻³ [1].	[3]
(c)	K _{sp} = [Ca ²⁺][OH ⁻] ² = (0.0174) × (0.0348) ² = 2.11 × 10 ⁻⁵ [2]; units: mol ³ dm ⁻⁹ [1].	[3]
(d)	CO ₂ reacts with OH ⁻ to form CaCO ₃ precipitate: Ca(OH) ₂ + CO ₂ → CaCO ₃ + H ₂ O [1]; reduces [OH ⁻] in solution, requiring smaller titre of HCl, leading to an underestimated K _{sp} [1].	[2]

MARK SCHEME: QUESTION 5 (PLANNING) — Planning Calorimetric Determination of the Hydration Enthalpy via Hess's Law**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Anhydrous $\text{MgCl}_2(\text{s}) + 6\text{H}_2\text{O}(\text{l}) \rightarrow \text{MgCl}_2 \cdot 6\text{H}_2\text{O}(\text{s})$ (H_{hyd}) [1]; $\text{MgCl}_2(\text{s}) + \text{aq} \rightarrow \text{MgCl}_2(\text{aq})$ (H_1) [1]; $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}(\text{s}) + \text{aq} \rightarrow \text{MgCl}_2(\text{aq})$ (H_2), correctly oriented with $H_{\text{hyd}} = H_1 - H_2$ [1].	[3]
(b)	50.0 cm ³ / 50.0 g of distilled water [1]; approximately 0.025 to 0.050 mol of solid (e.g. 2.4 g to 4.8 g anhydrous MgCl_2 , and stoichiometric equivalent 5.1 g to 10.2 g $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$) [1].	[2]
(c)	Record temperature of water every 30 seconds for 3 minutes before addition of solid [1]; add solid at minute 3.5, stir continuously, record temperature every 30 seconds from minute 4 to minute 10 [1]; plot temperature vs time and extrapolate cooling curve back to time of mixing (minute 3.5) to find theoretical maximum temperature T_{max} [1].	[3]
(d)	Anhydrous MgCl_2 is highly hygroscopic / deliquescent [1]; absorbs water vapor from air, causing error in calculated moles and releasing partial hydration heat before calorimetry [1].	[2]

MARK SCHEME: QUESTION 6 (ANALYSIS & EVALUATION) — Analysis and Evaluation of the Catalytic Decomposition of Hydrogen Peroxide**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	V_{∞} represents initial amount of H_2O_2 [1]; ($V_{\infty} - V_t$) is directly proportional to concentration of H_2O_2 remaining at time t [1].	[2]
(b)	Measure time taken for ($V_{\infty} - V_t$) to fall to half its initial value ($t_{1/2_1}$) [1]; then measure time taken to halve again ($t_{1/2_2}$); if $t_{1/2}$ is constant, the reaction is first-order [1].	[2]
(c)	Oxygen gas has slight solubility in aqueous solution [1]; gas leakage around the syringe plunger / delivery tube connection [1].	[2]
(d)	Doubling catalyst mass doubles available active surface area, doubling initial rate of reaction [1]; final volume V_{∞} remains unchanged because catalyst does not alter stoichiometry / moles of reactant [1].	[2]
(e)	Rapid filtration to remove solid catalyst / quenching by adding excess ice-cold sulfuric acid [2].	[2]

MARK SCHEME: QUESTION 7 (PLANNING) — Planning the Determination of Partition Coefficient (K_{pc}) of Ethanoic Acid**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Add measured volumes of aqueous ethanoic acid and octan-1-ol to separating funnel [1]; stopper, invert and shake vigorously for 5 minutes, periodically opening tap to release pressure, then clamp in retort stand and allow distinct layers to separate fully [1].	[2]
(b)	Remove stopper, open tap slowly and run out lower aqueous layer into clean beaker, discarding small boundary layer [2].	[2]
(c)	Pipette 10.0 cm ³ aliquot of each layer into separate conical flasks [1]; titrate with standardized 0.100 mol dm ⁻³ NaOH solution [1]; use phenolphthalein indicator (colourless to permanent pale pink endpoint) [1].	[3]
(d)	Wear nitrile gloves and eye protection (ethanoic acid is corrosive); handle organic solvent in fume cupboard away from naked flames (flammable vapour) [2].	[2]
(e)	Assumes ethanoic acid has the same molecular form (monomer) in both water and the organic solvent (does not dimerise significantly) [1].	[1]

MARK SCHEME: QUESTION 8 (ANALYSIS & EVALUATION) — Quantitative Analysis of Cell Potentials and Verification of the Nernst Equation**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	All log values correct: 0.00, -1.00, -2.00, -3.00, -4.00 [2].	[2]
(b)	Plot E_{cell} on y-axis against $\log[\text{Cu}^{2+}]$ on x-axis [2].	[2]
(c)	Theoretical gradient = $0.059 / 2 = +0.0295 \text{ V}$ (or $+0.030 \text{ V}$ per decade) [2].	[2]
(d)	E° is the y-intercept where $\log[\text{Cu}^{2+}] = 0.00$ (i.e. $[\text{Cu}^{2+}] = 1.00 \text{ mol dm}^{-3}$) [2].	[2]
(e)	Prevents any significant current from flowing through the cell [1]; ensures the concentrations of ions at the electrode surface remain unchanged (retains equilibrium conditions) [1].	[2]

MARK SCHEME: QUESTION 9 (PLANNING) — Planning the Determination of Molar Mass of a Volatile Liquid via the Dumas Method**[10 Marks]**

Part	Examiner Marking Points / Indicative Content	Marks
(a)	Self-sealing rubber septum fitted to syringe nozzle [1]; hypodermic micro-syringe used to puncture septum and inject volatile liquid [1].	[2]
(b)	Weigh micro-syringe containing hexane on balance (3 or 4 decimal places) [1]; inject liquid into gas syringe, re-weigh empty micro-syringe; mass injected = initial mass - final mass [1].	[2]
(c)	Maintain temperature well above boiling point of hexane (e.g. 85 °C to 95 °C) [1]; ensures hexane vaporises completely and behaves ideally as a gas [1].	[2]
(d)	1. Mass of injected liquid m (g); 2. Volume of vapor V (cm ³ / m ³); 3. Temperature of oven T (K); 4. Atmospheric pressure p (Pa) [2].	[2]
(e)	Intermolecular attractions cause the measured volume V to be smaller than ideal [1]; since $M \propto 1/V$, the calculated molar mass will be higher than true value [1].	[2]

MARK SCHEME: QUESTION 10 (ANALYSIS & EVALUATION) – Analysis of Apparatus Errors, Anomalies, and Standard Deviation in Calorimetry

[10 Marks]

Part	Examiner Marking Points / Indicative Content	Marks
(a)	% uncertainty = $(0.5 / 25.0) \times 100\% = 2.0\%$ [2].	[2]
(b)	Total reading error = $2 \times 0.1 = 0.2$ °C; % uncertainty = $(0.2 / 13.4) \times 100\% = 1.49\%$ (or 1.5%) [2].	[2]
(c)	Total % uncertainty = 2.0% (cylinder 1) + 2.0% (cylinder 2) + 1.49% (T) = 5.49% (or 5.5%) [2].	[2]
(d)	Anomaly: -58.1 kJ mol ⁻¹ [1]; Mean concordant = $(-54.2 - 53.9 - 54.0) / 3 = -54.03$ kJ mol ⁻¹ (accept -54.0 kJ mol ⁻¹) [1].	[2]
(e)	Use a volumetric pipette / burette instead of measuring cylinders for solution volumes [1]; use a vacuum Dewar flask / expanded polystyrene with lid to minimize heat loss to surroundings [1].	[2]